Storyline

Chapter 11: Angular Momentum

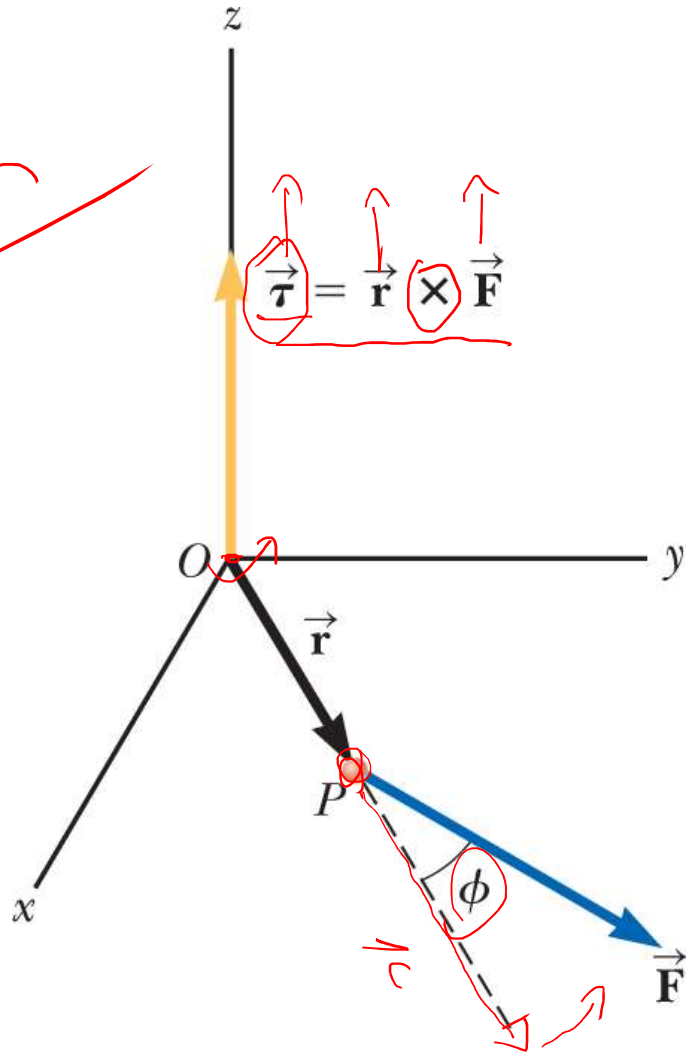
Physics for Scientists and Engineers, 10e
Raymond A. Serway
John W. Jewett, Jr.

BROOKS/COLE
CENGAGE Learning

The Vector Product and Torque

$$\vec{\boldsymbol{\tau}} \equiv \vec{\mathbf{r}} \times \vec{\mathbf{F}}$$

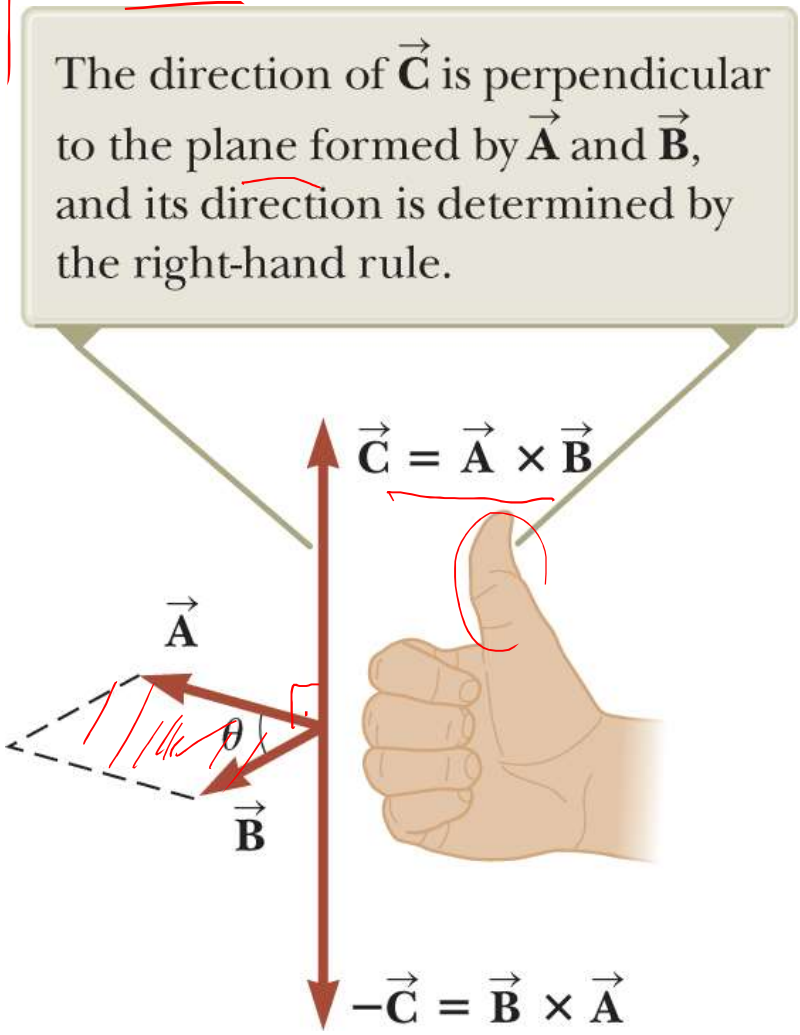
$$\left|\vec{\boldsymbol{\tau}}\right| = rF\sin\phi$$



The Vector Product and Torque

$$\vec{\mathbf{C}} = \vec{\mathbf{A}} \times \vec{\mathbf{B}}$$

$$C = AB \sin\theta$$



The Vector Product

Properties of vector product:

1. $\vec{\mathbf{A}} \times \vec{\mathbf{B}} = -\vec{\mathbf{B}} \times \vec{\mathbf{A}}$ → is not commutative

2. $\vec{\mathbf{A}} \times \vec{\mathbf{B}} = 0 \left(\vec{\mathbf{A}} \parallel \vec{\mathbf{B}}\right) \quad \Rightarrow \vec{\mathbf{A}} \times \vec{\mathbf{A}} = 0$

3. If $\vec{\mathbf{A}} \perp \vec{\mathbf{B}}$, then $\left|\vec{\mathbf{A}} \times \vec{\mathbf{B}}\right| = AB$

4. $\vec{\mathbf{A}} \times \left(\vec{\mathbf{B}} + \vec{\mathbf{C}}\right) = \vec{\mathbf{A}} \times \vec{\mathbf{B}} + \vec{\mathbf{A}} \times \vec{\mathbf{C}}$

5. $\dfrac{d}{dt}\left(\vec{\mathbf{A}} \times \vec{\mathbf{B}}\right) = \dfrac{d\vec{\mathbf{A}}}{dt} \times \vec{\mathbf{B}} + \vec{\mathbf{A}} \times \dfrac{d\vec{\mathbf{B}}}{dt}$

The Vector Product

$$\hat{\mathbf{i}}\times\hat{\mathbf{i}}=\hat{\mathbf{j}}\times\hat{\mathbf{j}}=\hat{\mathbf{k}}\times\hat{\mathbf{k}}=0$$

$$\hat{\mathbf{i}}\times\hat{\mathbf{j}}=-\hat{\mathbf{j}}\times\hat{\mathbf{i}}=\hat{\mathbf{k}}$$

$$\hat{\mathbf{j}}\times\hat{\mathbf{k}}=-\hat{\mathbf{k}}\times\hat{\mathbf{j}}=\hat{\mathbf{i}}$$

$$\hat{\mathbf{k}}\times\hat{\mathbf{i}}=-\hat{\mathbf{i}}\times\hat{\mathbf{k}}=\hat{\mathbf{j}}$$

$$\vec{\mathbf{A}}\times\left(-\vec{\mathbf{B}}\right)=-\vec{\mathbf{A}}\times\vec{\mathbf{B}}$$

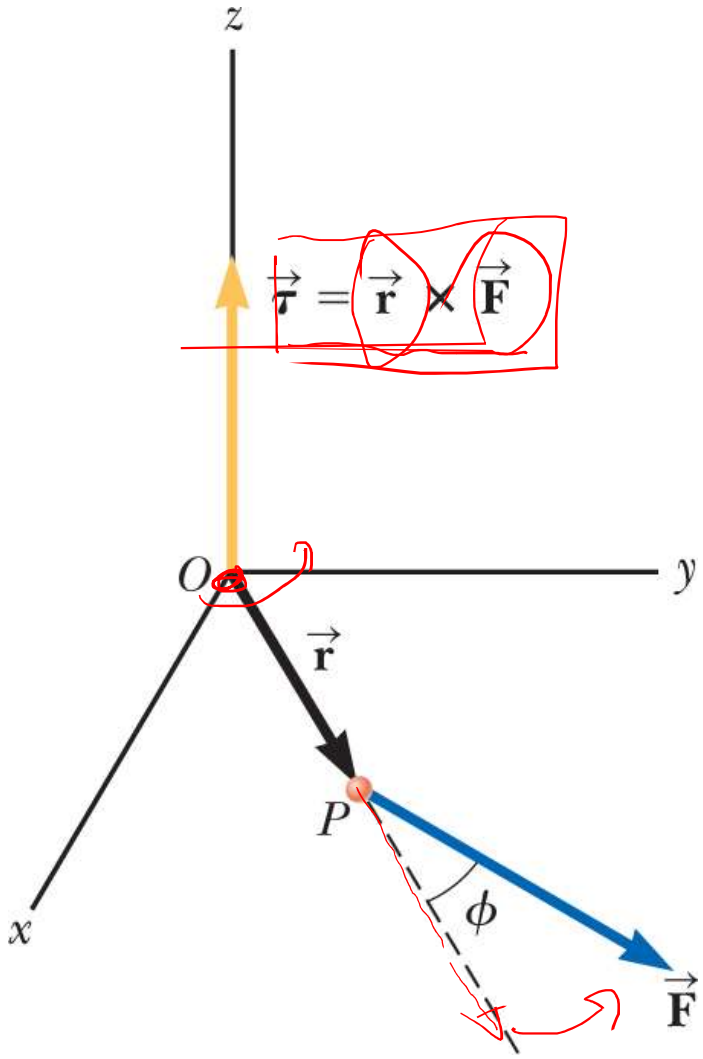
The Vector Product

$$\vec{\mathbf{A}} \times \vec{\mathbf{B}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$= \begin{vmatrix} A_y & A_z \\ B_y & B_z \end{vmatrix} \hat{\mathbf{i}} + \begin{vmatrix} A_z & A_x \\ B_z & B_x \end{vmatrix} \hat{\mathbf{j}} + \begin{vmatrix} A_x & A_y \\ B_x & B_y \end{vmatrix} \hat{\mathbf{k}}$$

$$\vec{\mathbf{A}} \times \vec{\mathbf{B}} = \left(A_y B_z - A_z B_y \right) \hat{\mathbf{i}} + \left(A_z B_x - A_x B_z \right) \hat{\mathbf{j}} + \left(A_x B_y - A_y B_x \right) \hat{\mathbf{k}}$$

Direction of Torque Vector



Quick Quiz 11.1

Which of the following statements about the relationship between the magnitude of the cross product of two vectors and the product of the magnitudes of the vectors is true?

(a) $\left|\vec{\mathbf{A}} \times \vec{\mathbf{B}}\right|$ is larger than AB.

(b) $\left|\vec{\mathbf{A}} \times \vec{\mathbf{B}}\right|$ is smaller than AB.

(c) $\left|\vec{\mathbf{A}} \times \vec{\mathbf{B}}\right|$ could be larger or smaller than AB, depending on the angle between the vectors.

(d) $\left|\vec{\mathbf{A}} \times \vec{\mathbf{B}}\right|$ could be equal to AB.

Quick Quiz 11.1

Which of the following statements about the relationship between the magnitude of the cross product of two vectors and the product of the magnitudes of the vectors is true?

(a) $\left|\vec{\mathbf{A}} \times \vec{\mathbf{B}}\right|$ is larger than AB.

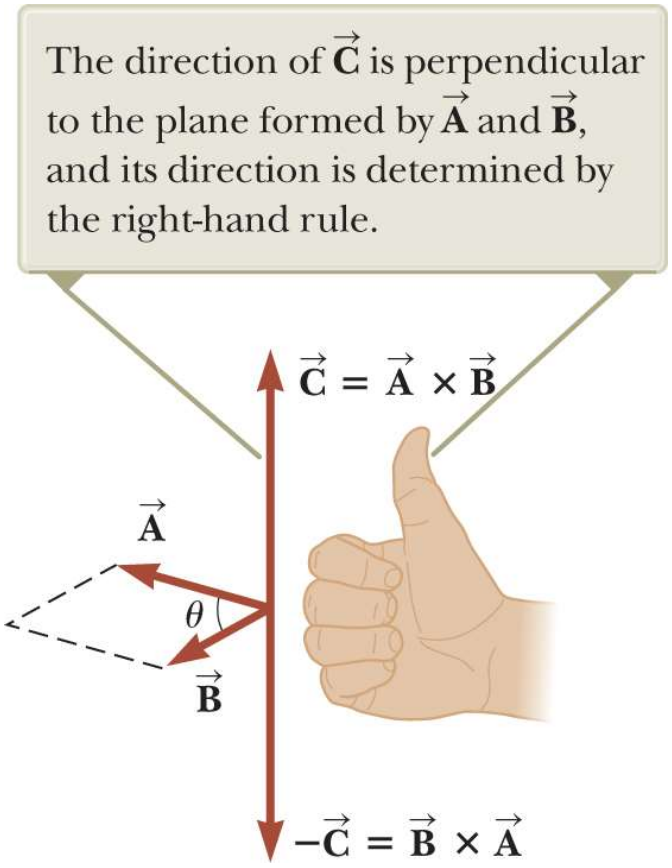
(b) $\left|\vec{\mathbf{A}} \times \vec{\mathbf{B}}\right|$ is smaller than AB.

(c) $\left|\vec{\mathbf{A}} \times \vec{\mathbf{B}}\right|$ could be larger or smaller than AB, depending on the angle between the vectors.

(d) $\left|\vec{\mathbf{A}} \times \vec{\mathbf{B}}\right|$ **could be equal to AB.**

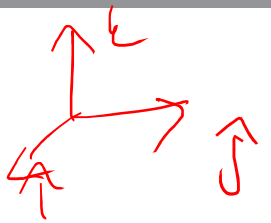
Example 11.1:
The Vector Product

Two vectors lying in the *xy* plane are given by the equations $\vec{\mathbf{A}} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}}$ and $\vec{\mathbf{B}} = -\hat{\mathbf{i}} + 2\hat{\mathbf{j}}$. Find $\vec{\mathbf{A}} \times \vec{\mathbf{B}}$ and verify that $\vec{\mathbf{A}} \times \vec{\mathbf{B}} = -\vec{\mathbf{B}} \times \vec{\mathbf{A}}$.



Example 11.1:
The Vector Product

$$\vec{\mathbf{A}} \times \vec{\mathbf{B}} = \left(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}}\right) \times \left(-\hat{\mathbf{i}} + 2\hat{\mathbf{j}}\right)$$



$$\vec{\mathbf{A}} \times \vec{\mathbf{B}} = 2\hat{\mathbf{i}} \times \left(-\hat{\mathbf{i}}\right) + 2\hat{\mathbf{i}} \times 2\hat{\mathbf{j}} + 3\hat{\mathbf{j}} \times \left(-\hat{\mathbf{i}}\right) + 3\hat{\mathbf{j}} \times 2\hat{\mathbf{j}}$$

$$\vec{\mathbf{A}} \times \vec{\mathbf{B}} = 0 + 4\hat{\mathbf{k}} + 3\hat{\mathbf{k}} + 0 = \boxed{7\hat{\mathbf{k}}}$$

$$\vec{\mathbf{B}} \times \vec{\mathbf{A}} = \left(-\hat{\mathbf{i}} + 2\hat{\mathbf{j}}\right) \times \left(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}}\right)$$

$$\vec{\mathbf{B}} \times \vec{\mathbf{A}} = \left(-\hat{\mathbf{i}}\right) \times 2\hat{\mathbf{j}} + \left(-\hat{\mathbf{i}}\right) \times 3\hat{\mathbf{j}} + 2\hat{\mathbf{j}} \times 2\hat{\mathbf{i}} + 2\hat{\mathbf{j}} \times 3\hat{\mathbf{j}}$$

$$\vec{\mathbf{B}} \times \vec{\mathbf{A}} = 0 - 3\hat{\mathbf{k}} - 4\hat{\mathbf{k}} + 0 = -7\hat{\mathbf{k}}$$

Example 11.2:
The Torque Vector

A force of $\vec{\mathbf{F}} = \left(2.00\hat{\mathbf{i}} + 3.00\hat{\mathbf{j}}\right)$ N is applied to a rigid object that is pivoted about a fixed axis aligned along the z coordinate axis. The force is applied at a point located at $\vec{\mathbf{r}} = \left(4.00\hat{\mathbf{i}} + 5.00\hat{\mathbf{j}}\right)$ m relative to the axis. Find the torque $\vec{\boldsymbol{\tau}}$ applied to the object.

Example 11.2:
The Torque Vector

$$\vec{\boldsymbol{\tau}} = \vec{\mathbf{r}} \times \vec{\mathbf{F}}$$

$$= \left[\left(4.00\hat{\mathbf{i}} + 5.00\hat{\mathbf{j}}\right)\ \mathrm{m}\right] \times \left[\left(2.00\hat{\mathbf{i}} + 3.00\hat{\mathbf{j}}\right)\ \mathrm{N}\right]$$

$$\vec{\boldsymbol{\tau}} = \left[(4.00)(2.00)\hat{\mathbf{i}} \times \hat{\mathbf{i}} + (4.00)(3.00)\hat{\mathbf{i}} \times \hat{\mathbf{j}} + (5.00)(2.00)\hat{\mathbf{j}} \times \hat{\mathbf{i}} + (5.00)(3.00)\hat{\mathbf{j}} \times \hat{\mathbf{j}}\right] \mathrm{N \cdot m}$$

$$\vec{\boldsymbol{\tau}} = \left[0 + 12.0\hat{\mathbf{k}} - 10.0\hat{\mathbf{k}} + 0\right] \mathrm{N \cdot m} = 2.0\hat{\mathbf{k}}\ \mathrm{N \cdot m}$$

Analysis Model: Nonisolated System (Angular Momentum)



Analysis Model: Nonisolated System (Angular Momentum)

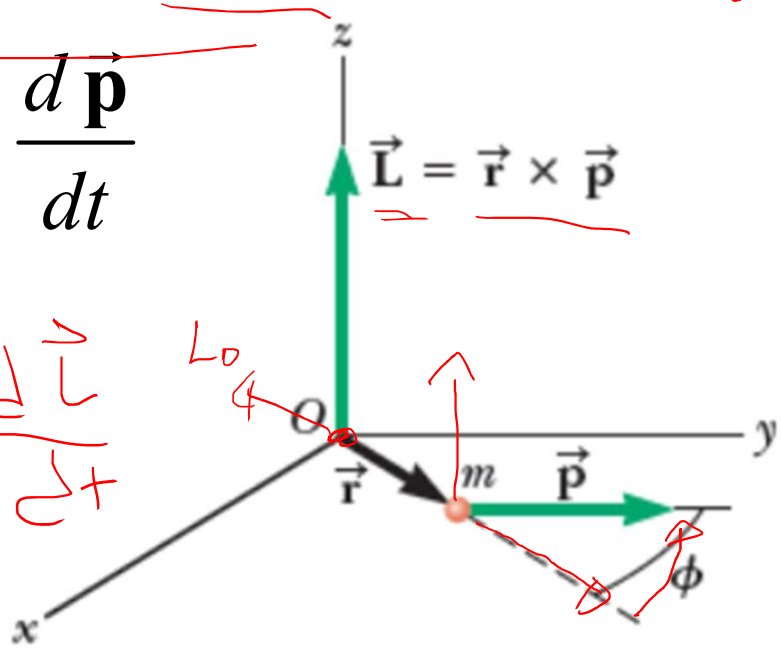
$$\sum \vec{\mathbf{F}}_{ext} = \frac{d\vec{\mathbf{p}}}{dt}$$

$$\vec{\mathbf{r}} \times \sum \vec{\mathbf{F}} = \sum \vec{\boldsymbol{\tau}} = \vec{\mathbf{r}} \times \frac{d\vec{\mathbf{p}}}{dt}$$

$$\frac{d}{dt}\left(\vec{\mathbf{r}} \times \vec{\mathbf{p}}\right) = \frac{d\vec{\mathbf{r}}}{dt} \times \vec{\mathbf{p}} + \vec{\mathbf{r}} \times \frac{d\vec{\mathbf{p}}}{dt}$$

$$\frac{d\vec{\mathbf{r}}}{dt} \times \vec{\mathbf{p}} = 0 \Rightarrow \frac{d}{dt}\left(\vec{\mathbf{r}} \times \vec{\mathbf{p}}\right) = \vec{\mathbf{r}} \times \frac{d\vec{\mathbf{p}}}{dt}$$

$$\sum \vec{\boldsymbol{\tau}}_{ext} = \frac{d\left(\vec{\mathbf{r}} \times \vec{\mathbf{p}}\right)}{dt} = \frac{d\vec{\mathbf{L}}}{dt}$$



Analysis Model: Nonisolated System (Angular Momentum)

The instantaneous **angular momentum** $\vec{\mathbf{L}}$ of a particle relative to an axis through a chosen origin O is defined by the cross product of the particle's instantaneous position vector $\vec{\mathbf{r}}$ relative to that origin and its instantaneous linear momentum $\vec{\mathbf{p}}$:

$$\vec{\mathbf{L}} \equiv \vec{\mathbf{r}} \times \vec{\mathbf{p}}$$

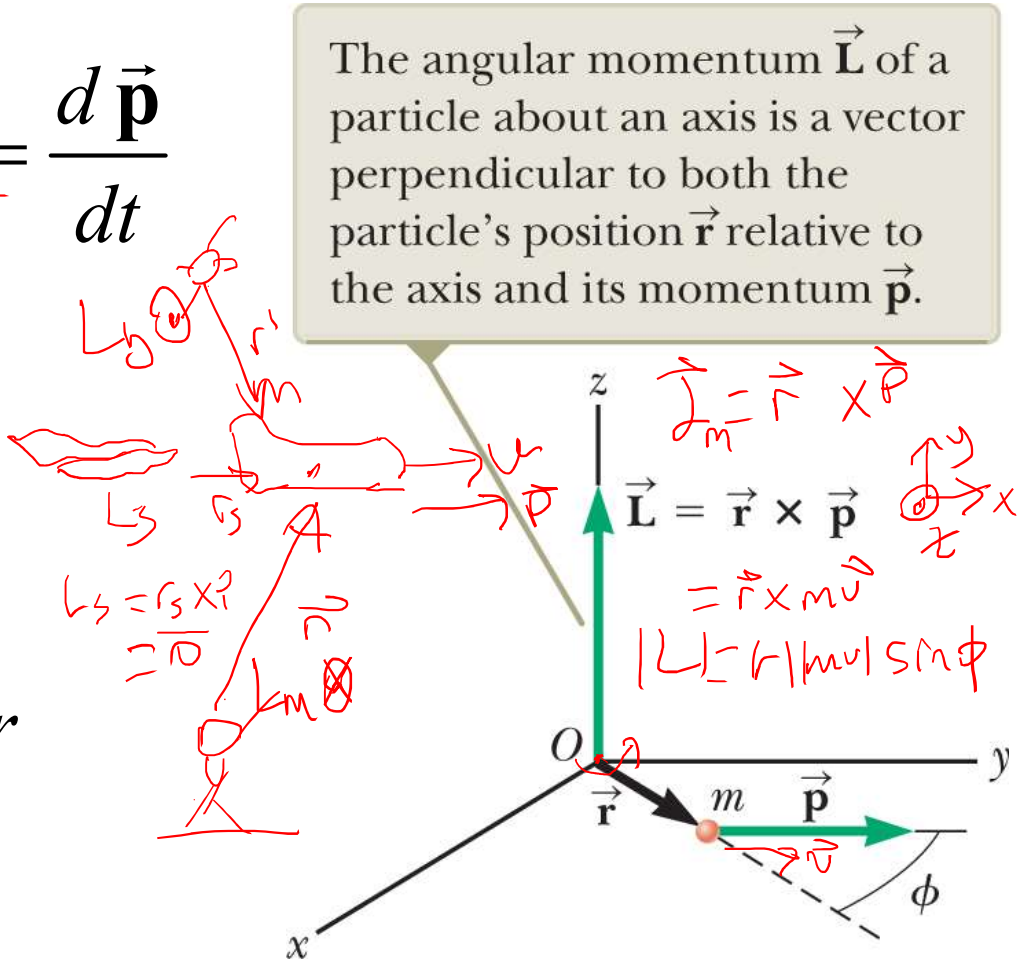
Analysis Model: Nonisolated System (Angular Momentum)

$$\sum \vec{\tau}_{ext} = \frac{d\vec{L}}{dt} \qquad \sum \vec{F}_{ext} = \frac{d\vec{p}}{dt}$$

$$L = mvr \sin\phi$$

$$\vec{r} \parallel \vec{p} \Rightarrow L = 0$$

$$\vec{r} \perp \vec{p} \Rightarrow L = mvr$$



Quick Quiz 11.2 Part I

A skater of mass m glides directly toward a pole with speed v. What is her angular momentum relative to the pole at the instant she is a distance d from the pole?

(a) zero
(b) mvd
(c) impossible to determine



Quick Quiz 11.2 Part I

A skater of mass m glides directly toward a pole with speed v. What is her angular momentum relative to the pole at the instant she is a distance d from the pole?

(a) zero

(b) mvd

(c) impossible to determine

Quick Quiz 11.2 Part II

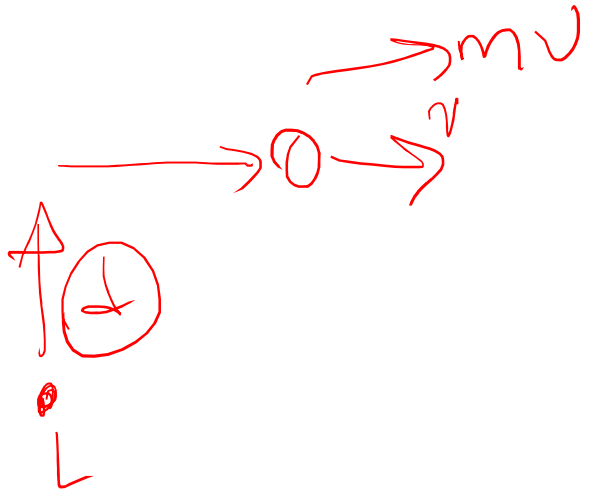
A skater of mass *m* glides at speed *v* along a straight path that is offset by a perpendicular distance *a* from a pole. What is her angular momentum relative to the pole?

(a) zero
(b) *mvd*
(c) *mva*
(d) impossible to determine

Quick Quiz 11.2 Part II

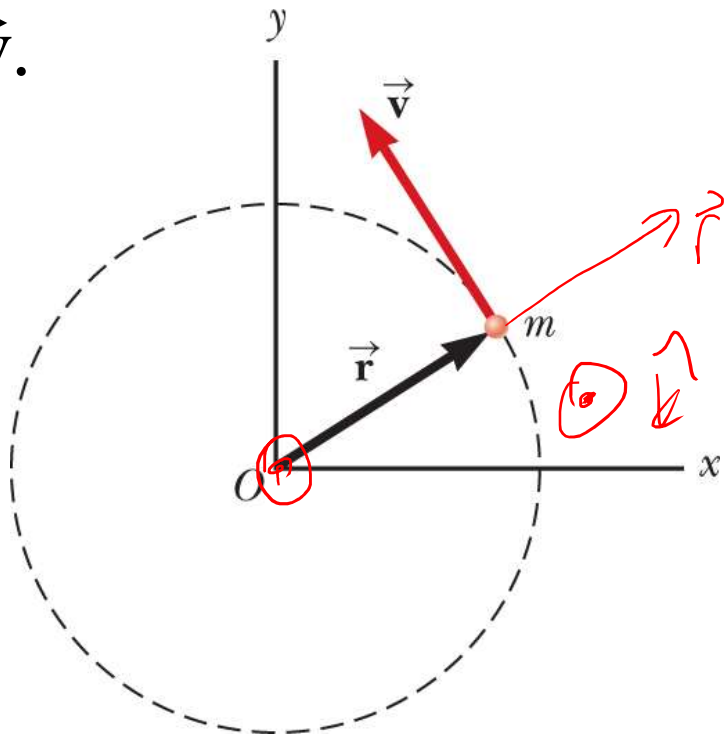
A skater of mass *m* glides at speed *v* along a straight path that is offset by a perpendicular distance *a* from a pole. What is her angular momentum relative to the pole?

(a) zero
(b) *mvd*
(c) *mva*
(d) impossible to determine



Example 11.3: Angular Momentum of a Particle in Uniform Circular Motion

A particle moves at constant speed in the xy plane in a circular path of radius r as shown in the figure. Find the magnitude and direction of its angular momentum relative to an axis through O when its velocity is $\vec{\mathbf{v}}$.



$$L = mvr\sin 90° = \boxed{mvr}$$

Angular Momentum of a System of Particles

$$\sum \vec{\mathbf{F}}_{\text{ext}} = \frac{d\,\vec{\mathbf{p}}_{\text{tot}}}{dt}$$

$$\vec{\mathbf{L}}_{\text{tot}} = \vec{\mathbf{L}}_1 + \vec{\mathbf{L}}_2 + \ldots + \vec{\mathbf{L}}_n = \sum_i \vec{\mathbf{L}}_i$$

$$\frac{d\,\vec{\mathbf{L}}_{\text{tot}}}{dt} = \sum_i \frac{d\,\vec{\mathbf{L}}_i}{dt} = \sum_i \vec{\boldsymbol{\tau}}_i$$

Angular Momentum of a System of Particles

External torque on a system

$$\sum \vec{\boldsymbol{\tau}}_{\text{ext}} = \frac{d\vec{\mathbf{L}}_{\text{tot}}}{dt}$$

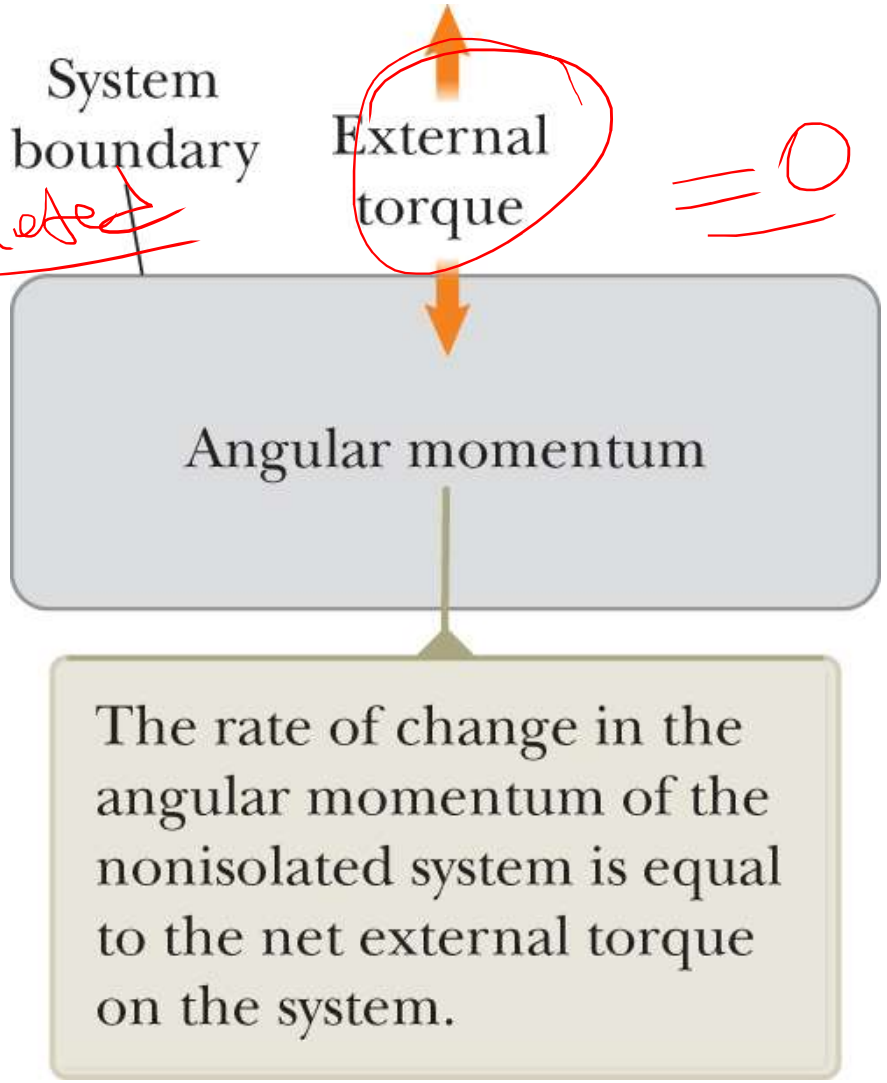
→ Lift Up

$$\sum \vec{\mathbf{F}}_{\text{ext}} = \frac{d\vec{\mathbf{p}}_{\text{tot}}}{dt}$$

$$\int \left(\sum \vec{\boldsymbol{\tau}}_{\text{ext}} \right) dt = \Delta \vec{\mathbf{L}}_{\text{tot}}$$

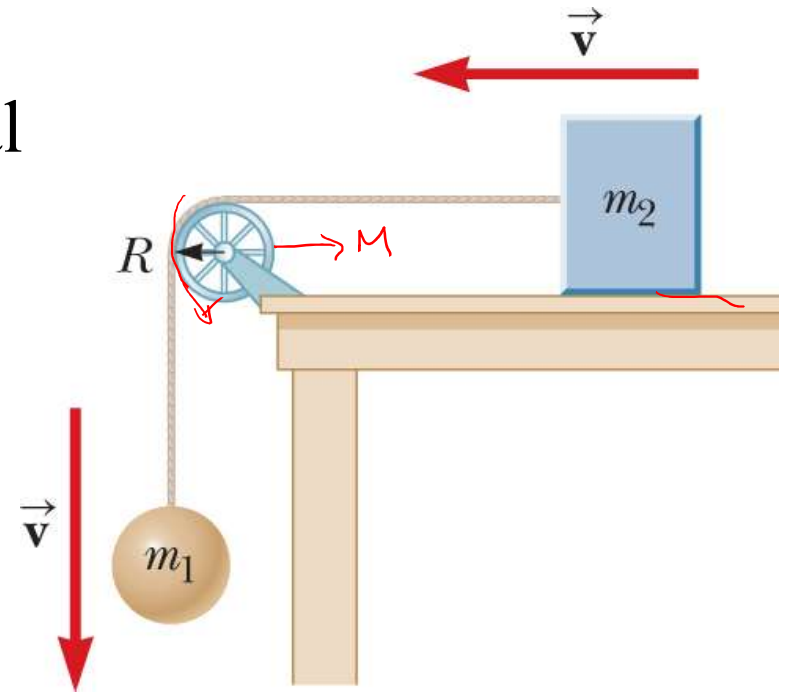
Analysis Model: Nonisolated System (Angular Momentum)

$$\sum \vec{\tau}_{\text{ext}} = \frac{d\vec{\mathbf{L}}_{\text{tot}}}{dt}$$



Example 11.4:
A System of Objects

A sphere of mass m_1 and a block of mass m_2 are connected by a light cord that passes over a pulley as shown in the figure. The radius of the pulley is R, and the mass of the thin rim is M. The spokes of the pulley have negligible mass. The block slides on a frictionless, horizontal surface. Find an expression for the linear acceleration of the two objects, using the concepts of angular momentum and torque.



Example 11.4: A System of Objects

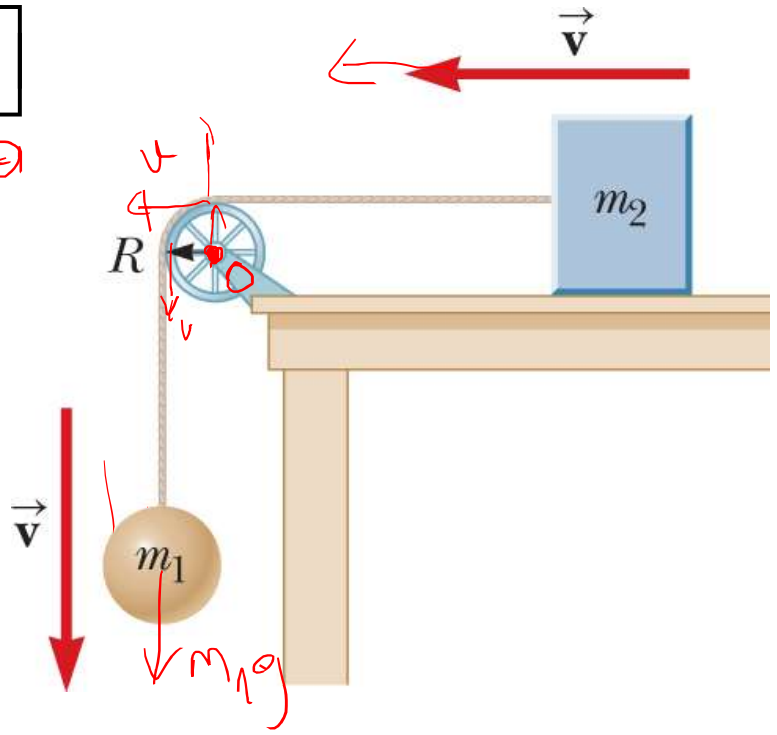
$$L = m_1 vR + m_2 vR + MvR = \left(m_1 + m_2 + M\right)vR$$

$$\sum \vec{\tau}_{\text{ext}} = \frac{d\vec{\mathbf{L}}_{\text{tot}}}{dt}$$

$$m_1 gR = \frac{d}{dt}\left[\left(m_1 + m_2 + M\right)vR\right]$$

$$m_1 gR = \left(m_1 + m_2 + M\right)R\frac{dv}{dt}$$

$$a = \boxed{\frac{m_1 g}{m_1 + m_2 + M}}$$



Angular Momentum of a Rotating Rigid Object

$$L_i = m_i r_i^2 \omega = m_i (r_i \omega) r_i = m_i r_i^2 \omega$$

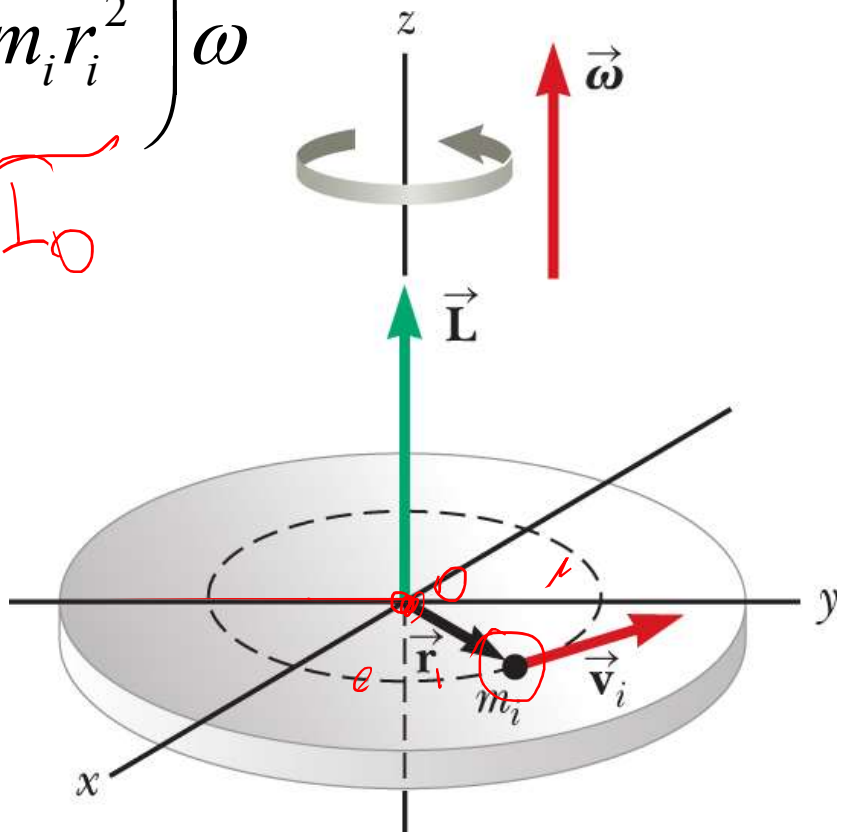
$$L_z = \sum_i L_i = \sum_i m_i r_i^2 \omega = \left(\sum_i m_i r_i^2 \right) \omega$$

$$L_z = I\omega$$

$$\frac{dL_z}{dt} = I \frac{d\omega}{dt} = I\alpha$$

$$\sum \tau_{\text{ext}} = I\alpha$$

$$\vec{\mathbf{L}} = I\vec{\boldsymbol{\omega}}$$



Quick Quiz 11.3

A solid sphere and a hollow sphere have the same mass and radius. They are rotating with the same angular speed. Which one has the higher angular momentum?

(a) the solid sphere

(b) the hollow sphere

(c) both have the same angular momentum

(d) impossible to determine

$I_{ss} = \frac{2}{5} M R^2$

$I_{hs} = \frac{MR^2}{2}$

$L = I \omega$

Quick Quiz 11.3

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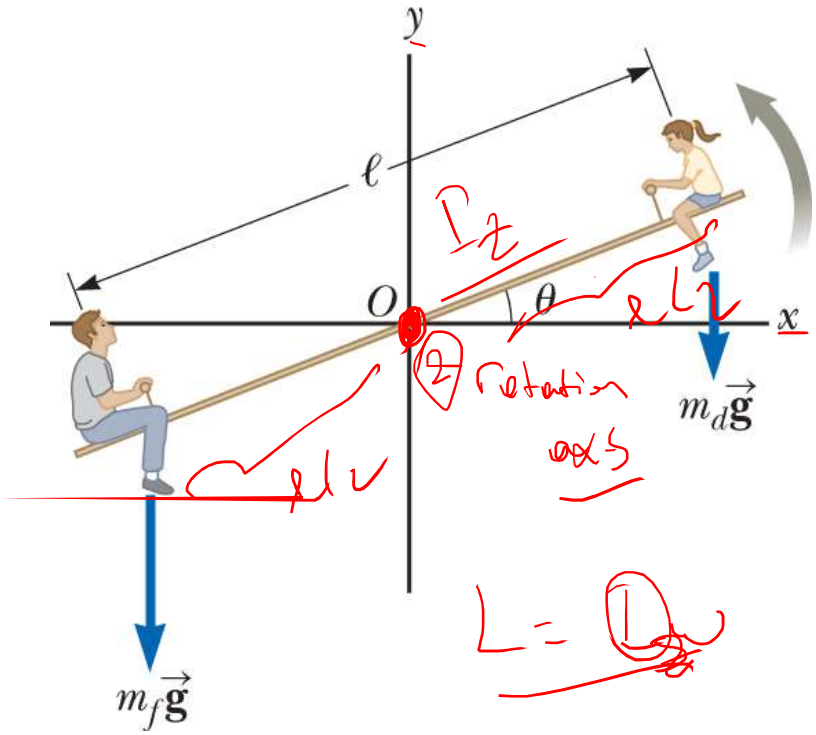
(c) both have the same angular momentum

(d) impossible to determine

Example 11.5: The Seesaw

A father of mass m_f and his daughter of mass m_d sit on opposite ends of a seesaw at equal distances from the pivot at the center. The seesaw is modeled as a rigid rod of mass M and length ℓ, and is pivoted without friction. At a given moment, the combination rotates in a vertical plane with an angular speed ω.

(A) Find an expression for the magnitude of the system's angular momentum.



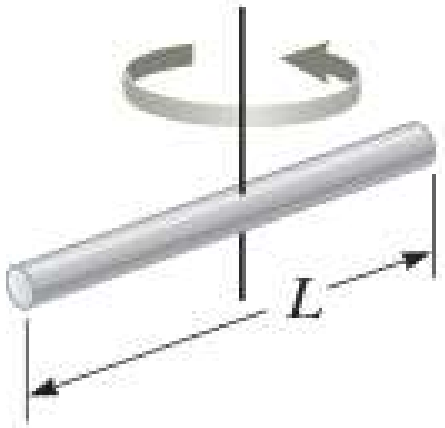
Example 11.5: The Seesaw

$$I = \frac{1}{12} M \ell^2 + m_f \left(\frac{\ell}{2}\right)^2 + m_d \left(\frac{\ell}{2}\right)^2 = \frac{\ell^2}{4}\left(\frac{M}{3} + m_f + m_d\right)$$

$$L = I\omega = \boxed{\frac{\ell^2}{4}\left(\frac{M}{3} + m_f + m_d\right)\omega}$$

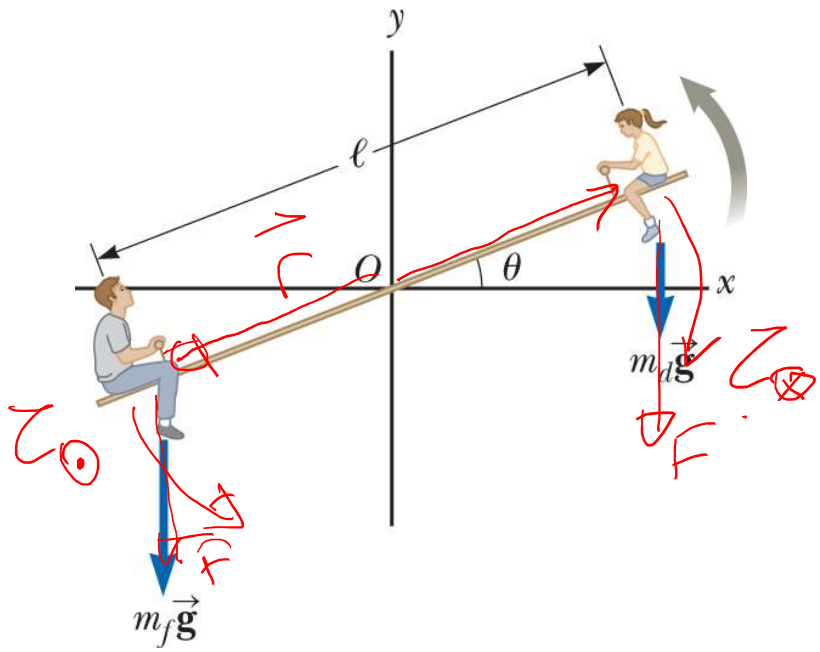
Long, thin rod with rotation axis through center

$$I_{CM} = \frac{1}{12} ML^2$$



Example 11.5:
The Seesaw

(B) Find an expression for the magnitude of the angular acceleration of the system when the seesaw makes an angle θ with the horizontal.



Example 11.5: The Seesaw

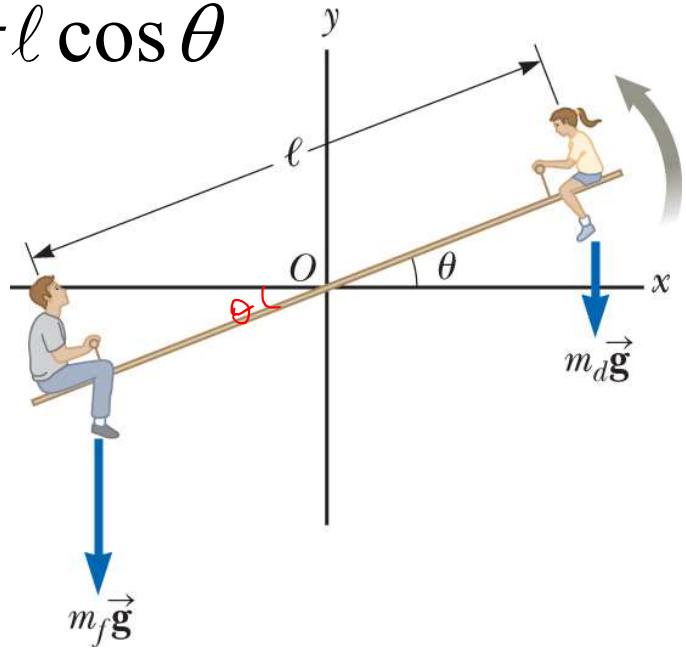
$$\tau_f = m_f g \frac{\ell}{2} \cos\theta \quad \left(\vec{\boldsymbol{\tau}}_f \text{ out of page}\right)$$

$$\tau_d = -m_d g \frac{\ell}{2} \cos\theta \quad \left(\vec{\boldsymbol{\tau}}_d \text{ into page}\right)$$

$$\sum \tau_{\text{ext}} = \tau_f + \tau_d = \frac{1}{2}\left(m_f - m_d\right) g \ell \cos\theta$$

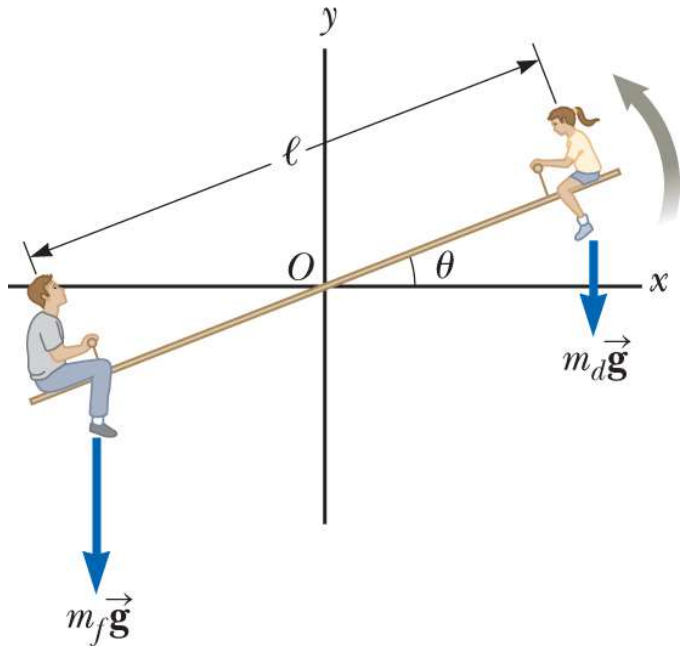
$$\alpha = \frac{\sum \tau_{\text{ext}}}{I}$$

$$\alpha = \frac{2\left(m_f - m_d\right) g \cos\theta}{\ell\left[\left(M/3\right) + m_f + m_d\right]}$$



Example 11.5:
The Seesaw

Imagine the father moves inward on the seesaw to a distance d from the pivot to try to balance the two sides. What is the angular acceleration of the system in this case when it is released from an arbitrary angle θ?

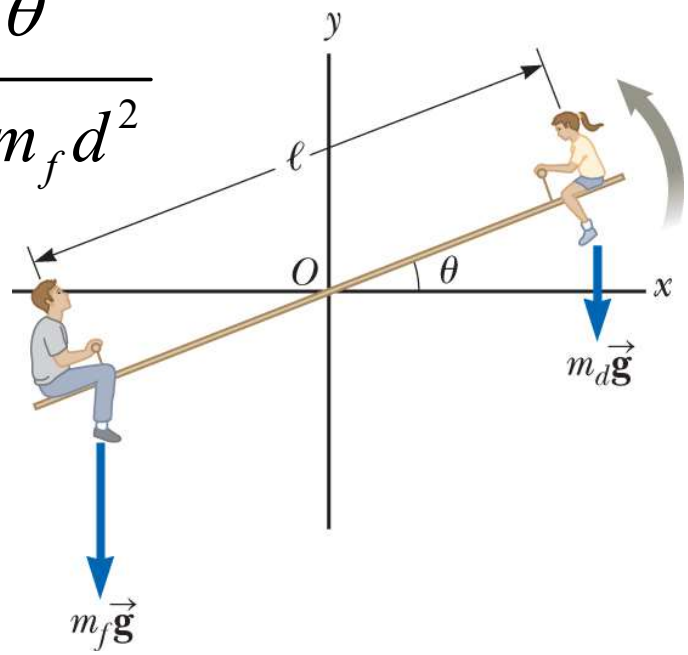


Example 11.5:
The Seesaw

$$I = \frac{1}{12} M \ell^2 + m_f d^2 + m_d \left(\frac{\ell}{2}\right)^2 = \frac{\ell^2}{4}\left(\frac{M}{3} + m_d\right) + m_f d^2$$

$$\sum \tau_{\text{ext}} = \tau_f + \tau_d = m_f g d \cos\theta - \frac{1}{2} m_d g \ell \cos\theta$$

$$\alpha = \frac{\sum \tau_{\text{ext}}}{I} = \frac{\left(m_f d - \frac{1}{2} m_d \ell\right) g \cos\theta}{\left(\ell^2/4\right)\left[\left(M/3\right) + m_d\right] + m_f d^2}$$

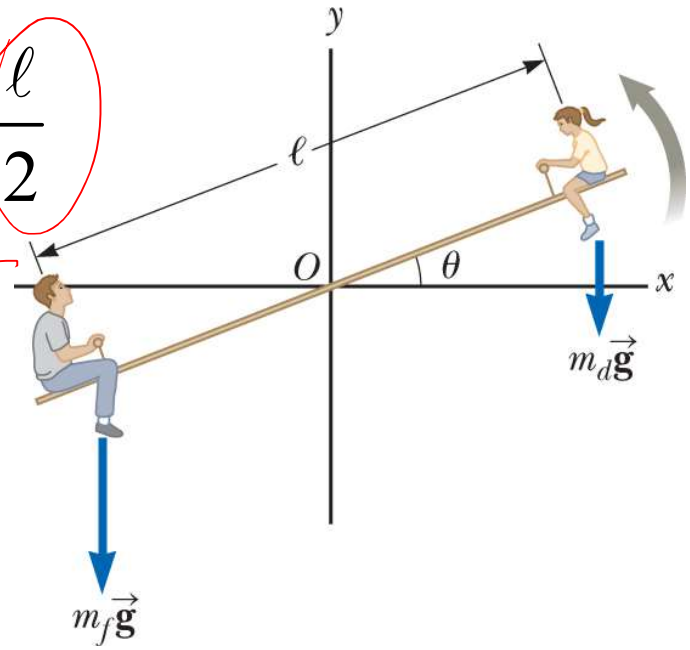


Example 11.5: The Seesaw

Where must the father sit for the seesaw to be balanced?

$$\alpha = \frac{\left(m_f d - \frac{1}{2} m_d \ell\right) g \cos\theta}{\left(\ell^2/4\right)\left[\left(M/3\right) + m_d\right] + m_f d^2} = 0$$

$$m_f d - \tfrac{1}{2} m_d \ell = 0 \rightarrow d = \left(\frac{m_d}{m_f}\right)\frac{\ell}{2}$$



Analysis Model: Isolated System (Angular Momentum)

The total angular momentum of a system is constant in both magnitude and direction if the net external torque acting on the system is zero, that is, if the system is isolated.

$$\sum \vec{\boldsymbol{\tau}}_{\text{ext}} = \frac{d\,\vec{\mathbf{L}}_{\text{tot}}}{dt} = 0$$

$$\Delta \vec{\mathbf{L}}_{\text{tot}} = 0$$

(conservation of angular momentum)

$$\vec{\mathbf{L}}_{\text{tot}} = \text{constant} \quad \text{or} \quad \vec{\mathbf{L}}_i = \vec{\mathbf{L}}_f$$

Analysis Model: Isolated System (Angular Momentum)

When his arms and legs are close to his body, the skater's moment of inertia is small and his angular speed is large.

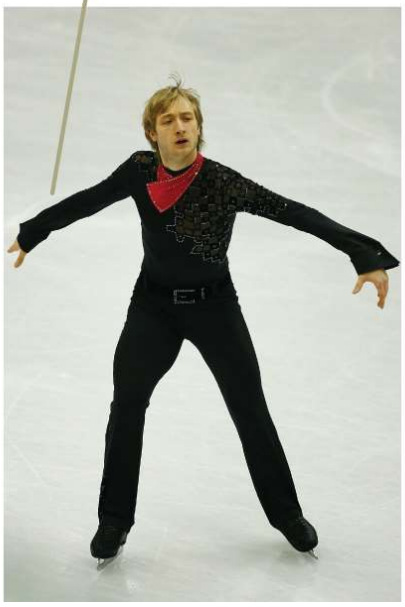


$$L = I\omega \text{ and}$$

$$\vec{\mathbf{L}}_{\text{tot}} = \text{constant}$$

$$I_i\omega_i = I_f\omega_f = \text{constant}$$

To slow down for the finish of his spin, the skater moves his arms and legs outward, increasing his moment of inertia.



Analysis Model: Isolated System (Angular Momentum)

$$\Delta E_{\text{system}} = 0 \quad (\text{no energy transfers across system boundary})$$

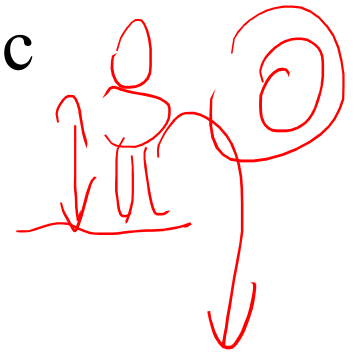
$$\Delta \vec{\mathbf{p}}_{\text{tot}} = 0 \quad (\text{net external force on system is zero})$$

$$\Delta \vec{\mathbf{L}}_{\text{tot}} = 0 \quad (\text{net external torque on system is zero})$$

Quick Quiz 11.4

A competitive diver leaves the diving board and falls toward the water with her body straight and rotating slowly. She pulls her arms and legs into a tight tuck position. What happens to her rotational kinetic energy?

(a) It increases.
(b) It decreases.
(c) It stays the same.
(d) It is impossible to determine.



Quick Quiz 11.4

A competitive diver leaves the diving board and falls toward the water with her body straight and rotating slowly. She pulls her arms and legs into a tight tuck position. What happens to her rotational kinetic energy?

(a) It increases.

(b) It decreases.

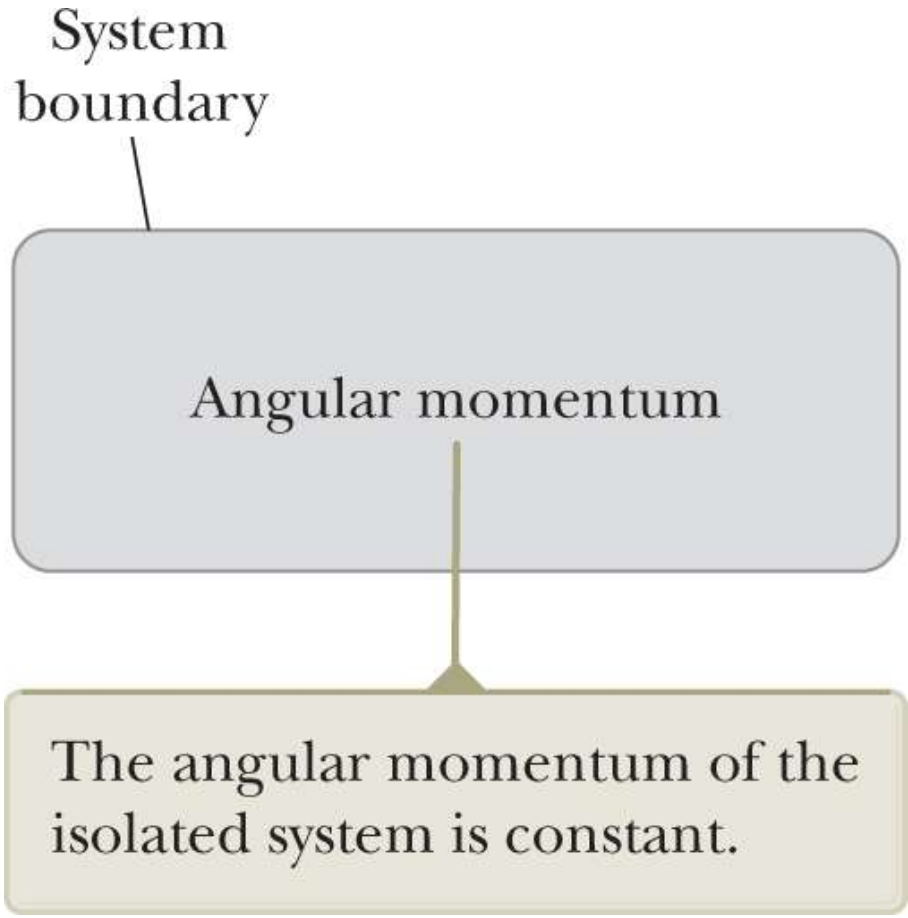
(c) It stays the same.

(d) It is impossible to determine.

Analysis Model: Isolated System (Angular Momentum)

$$\Delta \vec{\mathbf{L}}_{\text{tot}} = 0$$

$$I_i \omega_i = I_f \omega_f = \text{constant}$$



Example 11.6:
Formation of a Neutron Star

A star rotates with a period of 30 days about an axis through its center. The period is the time interval required for a point on the star's equator to make one complete revolution around the axis of rotation. After the star undergoes a supernova explosion, the stellar core, which had a radius of 10.0 km, collapses into a neutron star of radius 3.0 km. Determine the period of rotation of the neutron star.

Example 11.6:
Formation of a Neutron Star

$$I_i\omega_i = I_f\omega_f \qquad \Rightarrow I_i\left(\frac{2\pi}{T_i}\right) = I_f\left(\frac{2\pi}{T_f}\right)$$

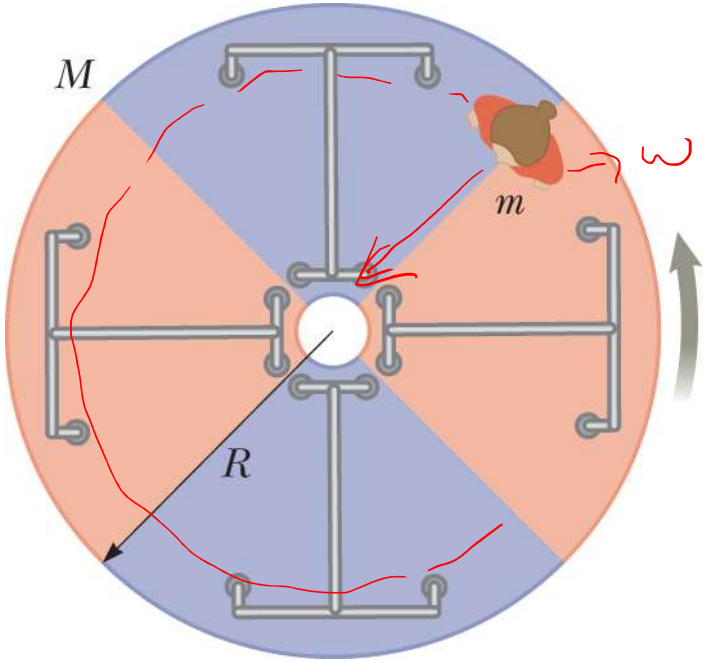
$$kMR_i^{\,2}\left(\frac{2\pi}{T_i}\right) = kMR_f^{\,2}\left(\frac{2\pi}{T_f}\right) \quad \Rightarrow T_f = \left(\frac{R_f}{R_i}\right)^2 T_i$$

$$T_f = \left(\frac{3.0\ \text{km}}{1.0\times10^4\ \text{km}}\right)^2 (30\ \text{days}) = 2.7\times10^{-6}\ \text{days}$$

$$= \boxed{0.23\ \text{s}}$$

Example 11.7:
The Merry-Go-Round

A horizontal platform in the shape of a circular disk rotates freely in a horizontal plane about a frictionless, vertical axle. The platform has a mass M = 100 kg and a radius R = 2.0 m. A student whose mass is m = 60 kg walks slowly from the rim of the disk toward its center. If the angular speed of the system is 2.0 rad/s when the student is at the rim, what is the angular speed when she reaches a point r = 0.50 m from the center?



Example 11.7: The Merry-Go-Round

$$I_i\omega_i = I_f\omega_f \qquad \Rightarrow \left(\frac{1}{2}MR^2 + mR^2\right)\omega_i = \left(\frac{1}{2}MR^2 + mr^2\right)\omega_f$$

$$\omega_f = \left(\frac{\frac{1}{2}MR^2 + mR^2}{\frac{1}{2}MR^2 + mr^2}\right)\omega_i$$

$$\omega_f = \left(\frac{\frac{1}{2}(100\text{ kg})(2.0\text{ m})^2 + (60\text{ kg})(2.0\text{ m})^2}{\frac{1}{2}(100\text{ kg})(2.0\text{ m})^2 + (60\text{ kg})(0.50\text{ m})^2}\right)(2.0\text{ rad/s})$$

$$= \left(\frac{440\text{ kg}\cdot\text{m}^2}{215\text{ kg}\cdot\text{m}^2}\right)(2.0\text{ rad/s}) = \boxed{4.1\text{ rad/s}}$$

Example 11.7:
The Merry-Go-Round

What if you measured the kinetic energy of the system before and after the student walks inward? Are the initial kinetic energy and the final kinetic energy the same?

$$K_i = \frac{1}{2} I_i \omega_i^2 = \frac{1}{2}\left(440 \text{ kg}\cdot\text{m}^2\right)\left(2.0 \text{ rad/s}\right)^2 = 880 \text{ J}$$

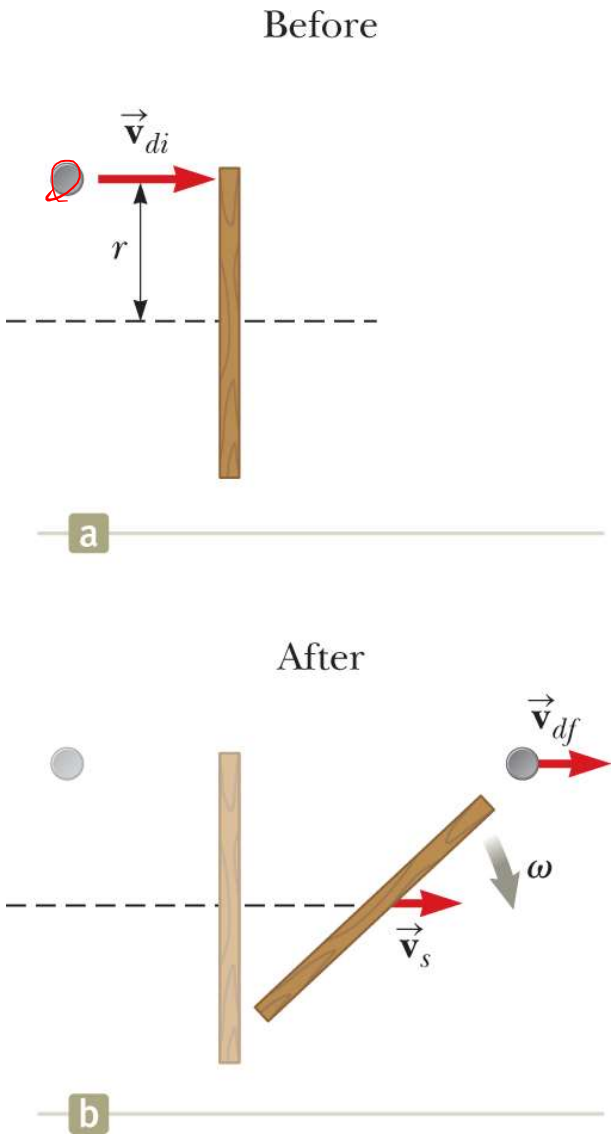
$$K_f = \frac{1}{2} I_f \omega_f^2 = \frac{1}{2}\left(215 \text{ kg}\cdot\text{m}^2\right)\left(4.1 \text{ rad/s}\right)^2 = 1.80\times10^3 \text{ J}$$

Kinetic energy increases

Example 11.8:
Disk and Stick Collision

A 2.0-kg disk traveling at 3.0 m/s strikes a 1.0-kg stick of length 4.0 m that is lying flat on nearly frictionless ice as shown in the overhead view of the top figure. The disk strikes at the endpoint of the stick, at a distance $r = 2.0$ m from the stick's center. Assume the collision is elastic and the disk does not deviate from its original line of motion.

Find the translational speed of the disk, the translational speed of the stick, and the angular speed of the stick after the collision. The moment of inertia of the stick about its center of mass is 1.33 kg · m^2.



Example 11.9: Disk and Stick Collision

$$\Delta\vec{\mathbf{p}}_{\text{tot}} = 0 \rightarrow \left(m_d v_{df} + m_s v_s\right) - m_d v_{di} = 0$$

$$(1)\ m_d\left(v_{di} - v_{df}\right) = m_s v_s$$

$$\Delta\vec{\mathbf{L}}_{\text{tot}} = 0 \rightarrow \left(-r m_d v_{df} + I\omega\right) - \left(-r m_d v_{di}\right) = 0$$

$$(2)\ -r m_d\left(v_{di} - v_{df}\right) = I\omega$$

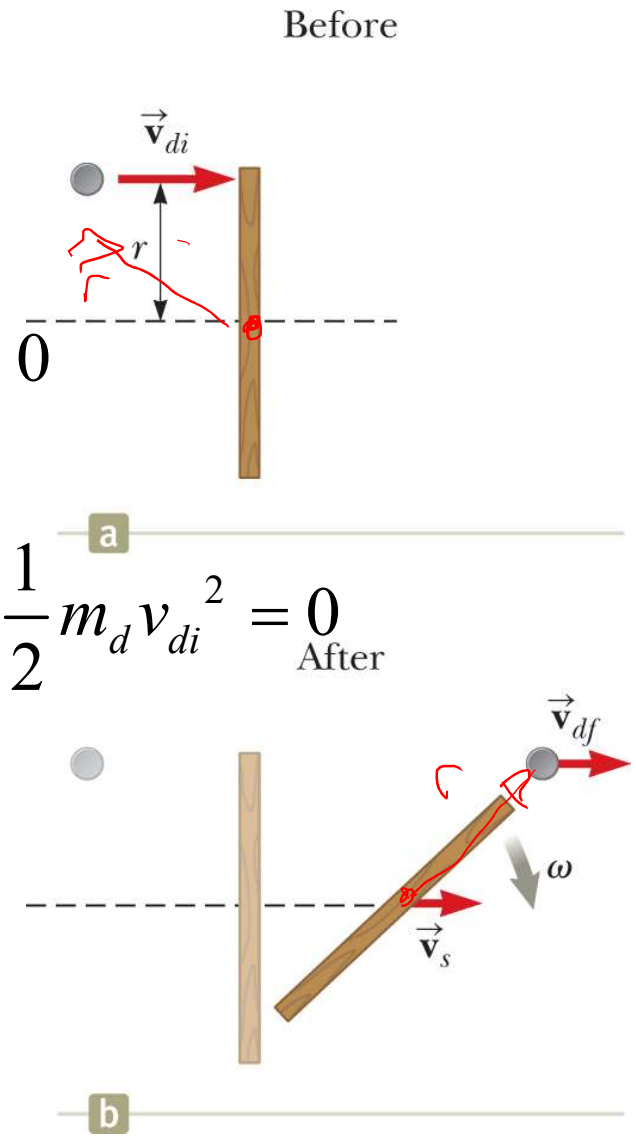
$$\Delta K = 0 \rightarrow \left(\frac{1}{2} m_d v_{df}^{\,2} + \frac{1}{2} m_s v_s^{\,2} + \frac{1}{2} I\omega^2\right) - \frac{1}{2} m_d v_{di}^{\,2} = 0$$

$$(3)\ m_d\left(v_{di} - v_{df}\right)\left(v_{di} + v_{df}\right) = m_s v_s^{\,2} + I\omega^2$$

$$r m_d\left(v_{di} - v_{df}\right) = r m_s v_s$$

$$-r m_d\left(v_{di} - v_{df}\right) = I\omega$$

$$0 = r m_s v_s + I\omega$$



Example 11.8: Disk and Stick Collision

$$0 = rm_s v_s + I\omega \rightarrow (4)\ \omega = -\frac{rm_s v_s}{I}$$

Eq(3) / Eq(1)

$$\frac{m_d\left(v_{di} - v_{df}\right)\left(v_{di} + v_{df}\right)}{m_d\left(v_{di} - v_{df}\right)} = \frac{m_s v_s^2 + I\omega^2}{m_s v_s}$$

$$(5)\ v_{di} + v_{df} = v_s + \frac{I\omega^2}{m_s v_s}$$

$$\omega = -\frac{rm_s v_s}{I} \rightarrow (6)\ v_{di} + v_{df} = v_s\left(1 + \frac{r^2 m_s}{I}\right)$$

#1 $$m_d\left(v_{di} - v_{df}\right) = m_s v_s \rightarrow v_{df} = v_{di} - \frac{m_s}{m_d} v_s$$

$$\rightarrow v_{di} + \left(v_{di} - \frac{m_s}{m_d} v_s\right) = v_s\left(1 + \frac{r^2 m_s}{I}\right)$$

Example 11.8:
Disk and Stick Collision

$$v_{di} + \left(v_{di} - \frac{m_s}{m_d} v_s \right) = v_s \left(1 + \frac{r^2 m_s}{I} \right) \rightarrow v_s = \frac{2 v_{di}}{1 + \left(m_s / m_d \right) + \left(r^2 m_s / I \right)}$$

$$v_s = \frac{2\left(3.0 \text{ m/s}\right)}{1 + \left(1.0 \text{ kg}/2.0 \text{ kg}\right) + \left[\left(2.0 \text{ m}\right)^2 \left(1.0 \text{ kg}\right) / 1.33 \text{ kg} \cdot \text{m}^2 \right]} = \boxed{1.3 \text{ m/s}}$$

$$\omega = -\frac{r m_s v_s}{I} = -\frac{\left(2.0 \text{ m}\right)\left(1.0 \text{ kg}\right)\left(1.3 \text{ m/s}\right)}{1.33 \text{ kg} \cdot \text{m}^2} = \boxed{-2.0 \text{ rad/s}}$$

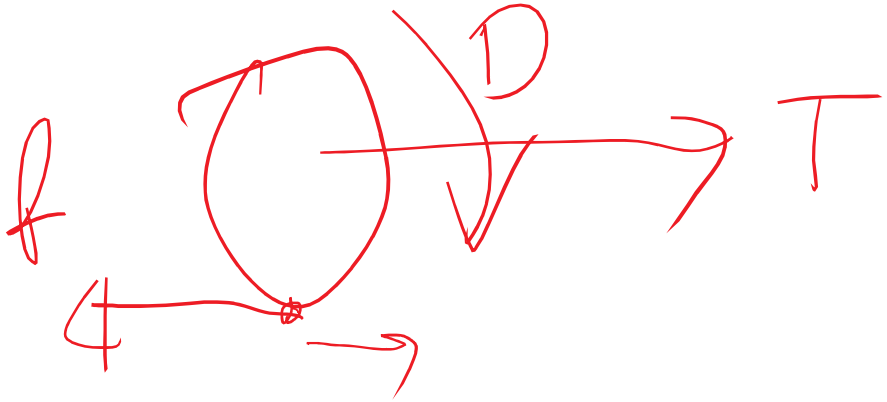
$$v_{df} = v_{di} - \frac{m_s}{m_d} v_s = 3.0 \text{ m/s} - \frac{1.0 \text{ kg}}{2.0 \text{ kg}} \left(1.3 \text{ m/s}\right) = \boxed{2.3 \text{ m/s}}$$

Example 11.8: Disk and Stick Collision

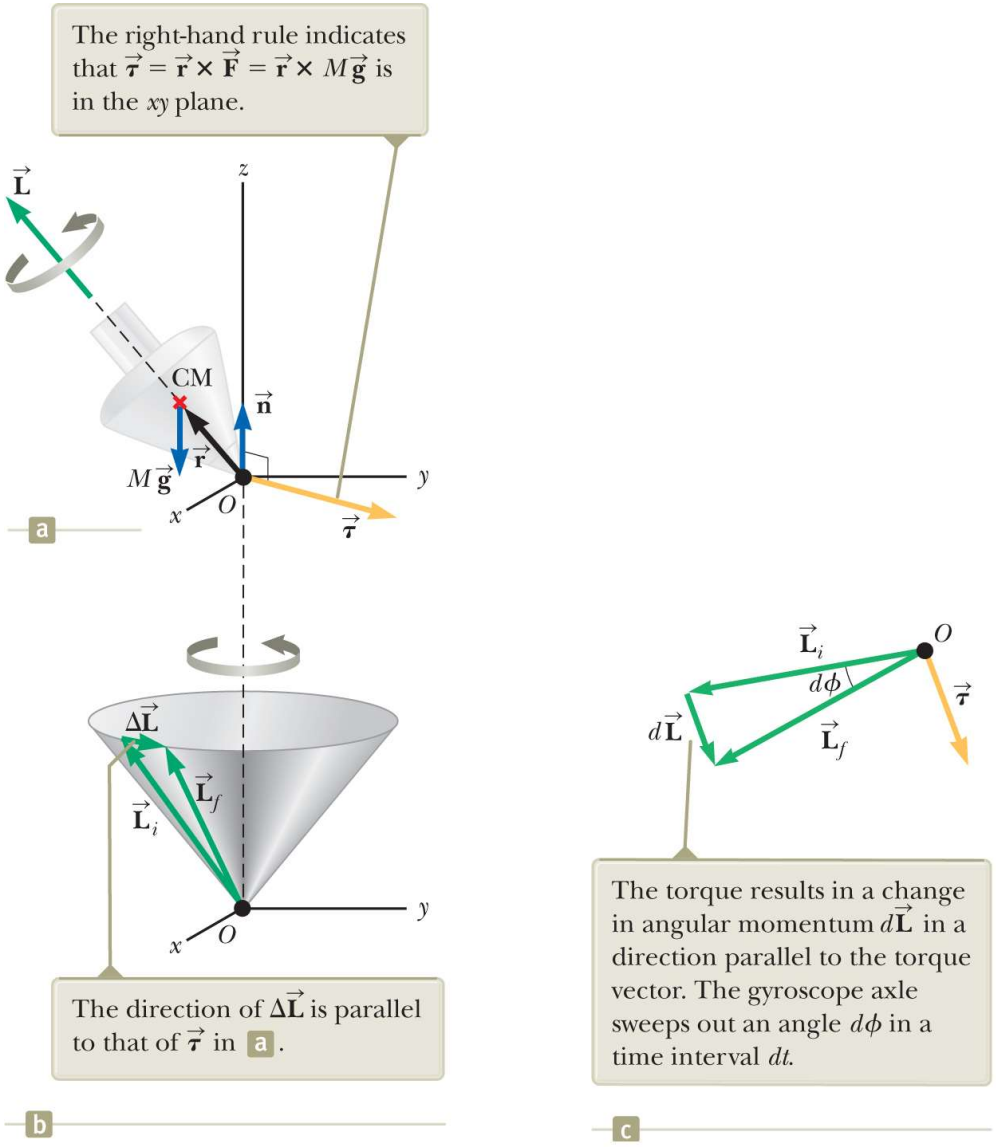
TABLE 11.1 Comparison of Values in Example 11.8 Before and After the Collision

	v (m/s)	ω (rad/s)	p (kg · m/s)	L (kg · m^2/s)	K_{trans} (J)	K_{rot} (J)
Before						
Disk	3.0	—	6.0	−12	9.0	—
Stick	0	0	0	0	0	0
Total for system	—	—	6.0	−12	9.0	0
After						
Disk	2.3	—	4.7	−9.3	5.4	—
Stick	1.3	−2.0	1.3	−2.7	0.9	2.7
Total for system	—	—	6.0	−12	6.3	2.7

Note: Linear momentum, angular momentum, and total kinetic energy of the system are all conserved.



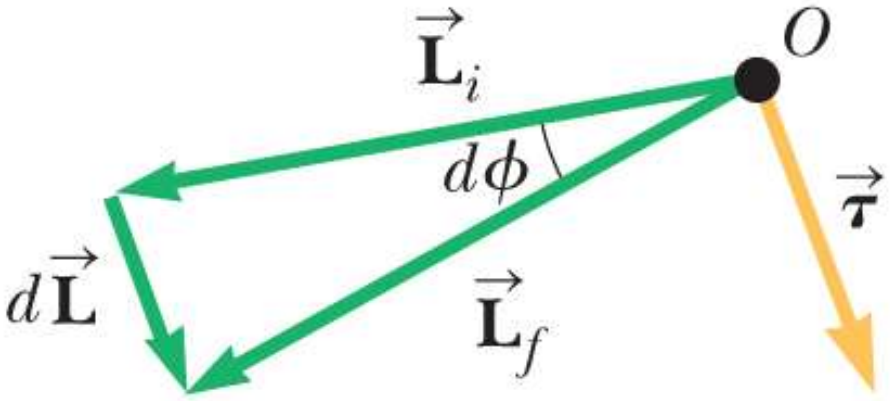
The Motion of Gyroscopes and Tops



$$\sum \vec{\tau}_{\text{ext}} = \frac{d\vec{\mathbf{L}}}{dt}$$

$$d\vec{\mathbf{L}} = \vec{\mathbf{L}}_f - \vec{\mathbf{L}}_i = \vec{\tau}\, dt$$

The Motion of Gyroscopes and Tops



$$d\phi = \frac{dL}{L} = \frac{\sum \tau_{\text{ext}} dt}{L} = \frac{\left(Mgr_{\text{CM}}\right) dt}{L}$$

$$\omega_p = \frac{d\phi}{dt} = \frac{Mgr_{\text{CM}}}{I\omega}$$

Assessing to Learn

Which situation has the least magnitude of angular momentum about the origin?

A. A 2-kg mass travels along the line $y = 3$ m with speed 1.5 m/s.

B. A 1-kg mass travels in a circle of $r = 4.5$ m about the origin with speed 2 m/s.

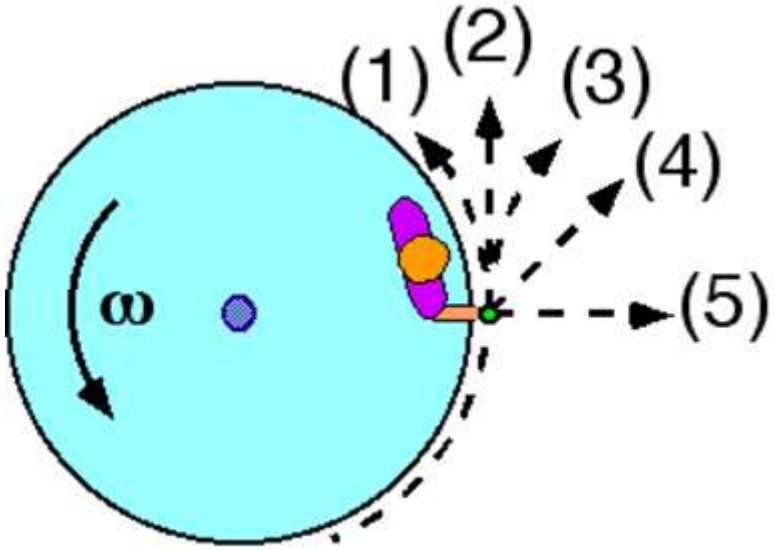
C. A disk with $I = 3$ kg·m^2 rotates about its center (on origin) with $\omega = 3$ rad/s.

1. A
2. B
3. C
4. Both A and B
5. Both A and C
6. Both B and C
7. All have the same magnitude angular momentum

Assessing to Learn

A child is standing at the rim of a rotating disk holding a rock. The disk rotates freely without friction. If the rock is dropped at the instant shown, which of the indicated paths most nearly represents the path of the rock as seen from above the disk?

1. path (1)
2. path (2)
3. path (3)
4. path (4)
5. path (5)
6. cannot be determined

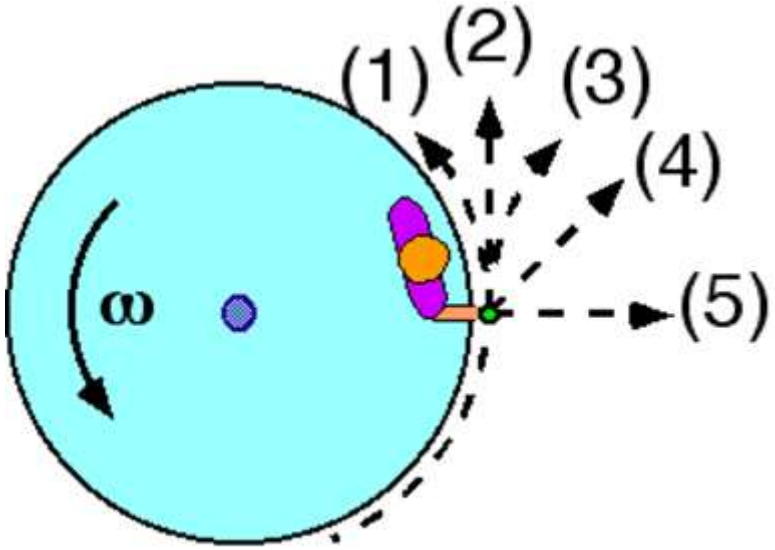


View from above

Assessing to Learn

A child is standing at the rim of a freely rotating disk holding a rock. The disk rotates without friction. The rock is dropped at the instant shown. As a result of dropping the rock, what happens to the angular velocity of the child and disk?

1. increases
2. stays the same
3. decreases
4. cannot be determined

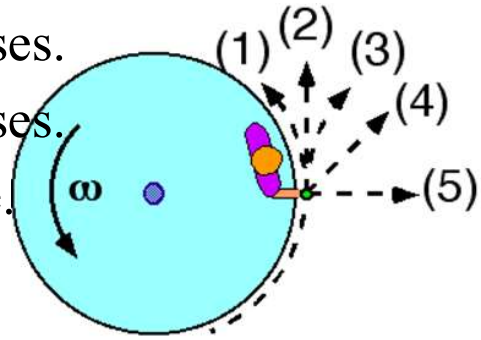


View from above

Assessing to Learn

A child is standing at the rim of a freely rotating disk holding a rock. The disk rotates without friction. The rock is dropped at the instant shown. Which of the following statements is true about the process of dropping the rock?

1. Angular momentum is conserved, mechanical energy increases.
2. Angular momentum is conserved, mechanical energy decreases.
3. Angular momentum increases, mechanical energy is conserved.
4. Angular momentum decreases, mechanical energy is conserved.
5. Both angular momentum and mechanical energy are conserved.
6. Both angular momentum and mechanical energy increase.
7. Angular momentum decreases, mechanical energy increases.
8. Angular momentum increases, mechanical energy decreases.
9. Both angular momentum and mechanical energy decrease.
10. The conserved quantities cannot be determined.

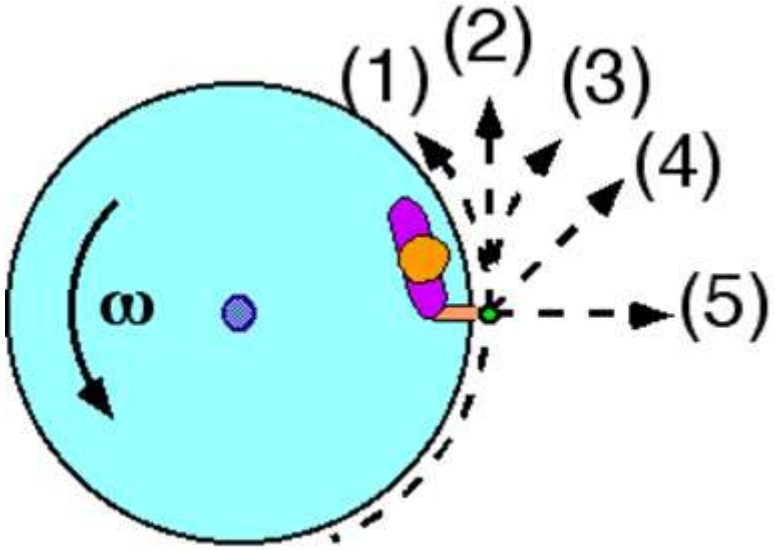


View from above

Assessing to Learn

A child is standing at the rim of a disk holding a rock. The disk rotates freely without friction. At the instant shown, the child throws the rock *radially outward*. Which of the indicated paths most nearly represents the trajectory of the rock as seen from above?

1. path (1)
2. path (2)
3. path (3)
4. path (4)
5. path (5)
6. none of the above
7. cannot be determined

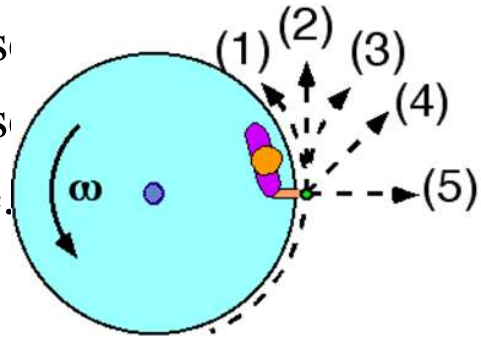


View from above

Assessing to Learn

A child is standing at the rim of a rotating disk, and throws a rock *radially outward* at the instant shown. The disk rotates freely without friction. Which of the following statements is correct about the disk-child-rock system as the rock is thrown?

1. Angular momentum is conserved, mechanical energy increases.
2. Angular momentum is conserved, mechanical energy decreases.
3. Angular momentum increases, mechanical energy is conserved.
4. Angular momentum decreases, mechanical energy is conserved.
5. Both angular momentum and mechanical energy are conserved.
6. Both angular momentum and mechanical energy increase.
7. Angular momentum decreases, mechanical energy increas
8. Angular momentum increases, mechanical energy decreas
9. Both angular momentum and mechanical energy decrease
10. The conserved quantities cannot be determined.

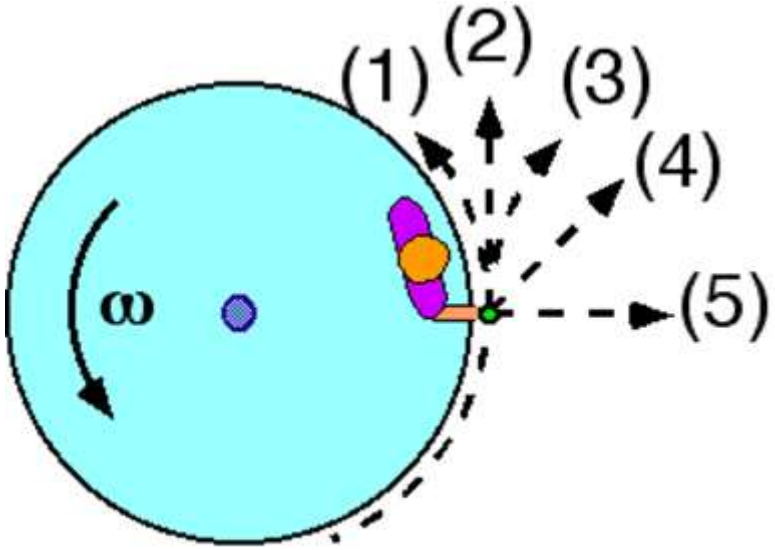


View from above

Assessing to Learn

A child is standing at the rim of a rotating disk holding a rock, and throws a rock in direction (2) at the instant shown. The disk rotates freely without friction. What happens to the angular speed of the disk?

1. increases
2. remains the same
3. decreases
4. impossible to determine



View from above

Assessing to Learn

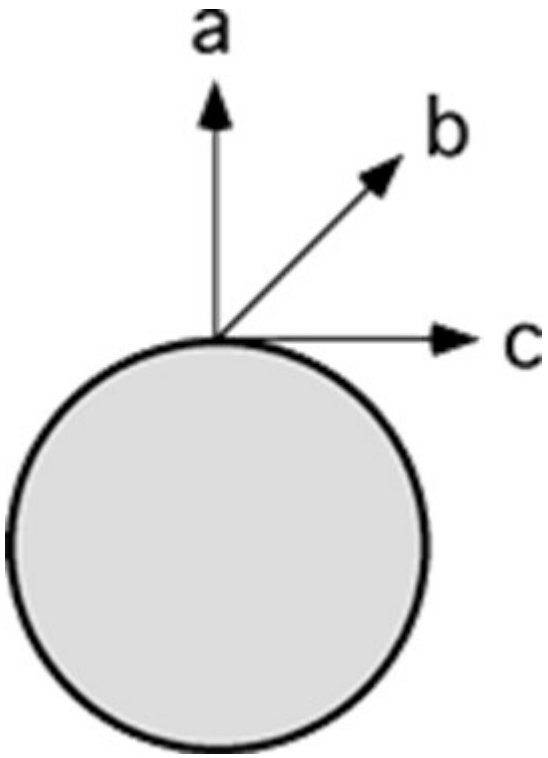
An ice skater begins a spin in the middle of a large rink, but then starts to spin faster by pulling her arms in. Which of the following statements is true?

1. Both kinetic energy and angular momentum are conserved.
2. Kinetic energy is conserved; angular momentum increases.
3. Kinetic energy is conserved; angular momentum decreases.
4. Kinetic energy increases; angular momentum is conserved.
5. Kinetic energy decreases; angular momentum is conserved.
6. Both kinetic energy and angular momentum increase.
7. Kinetic energy increases; angular momentum decreases.
8. Kinetic energy decreases; angular momentum increases.
9. Both kinetic energy and angular momentum decrease.
10. Impossible to determine.

Assessing to Learn

Two masses m_1 and m_2, having $m_1 > m_2$, are launched with the same speed in the direction a. Which mass reaches the greatest height?

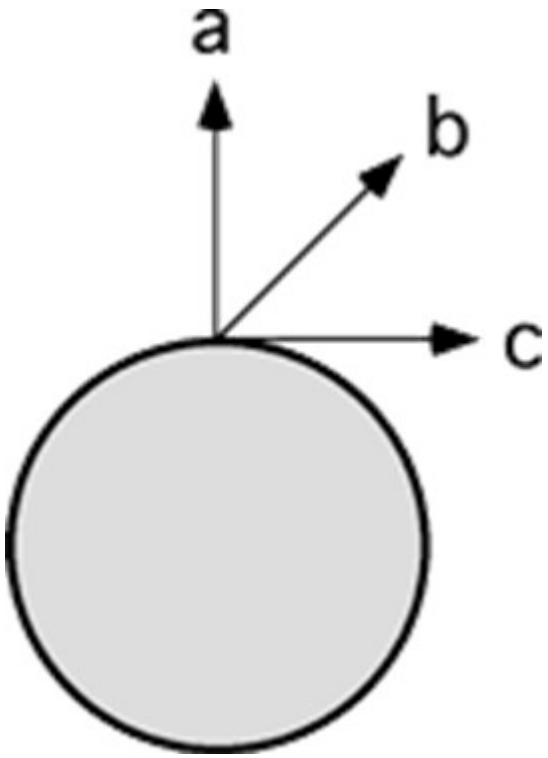
1. m_1
2. m_2
3. Both reach the same height.



Assessing to Learn

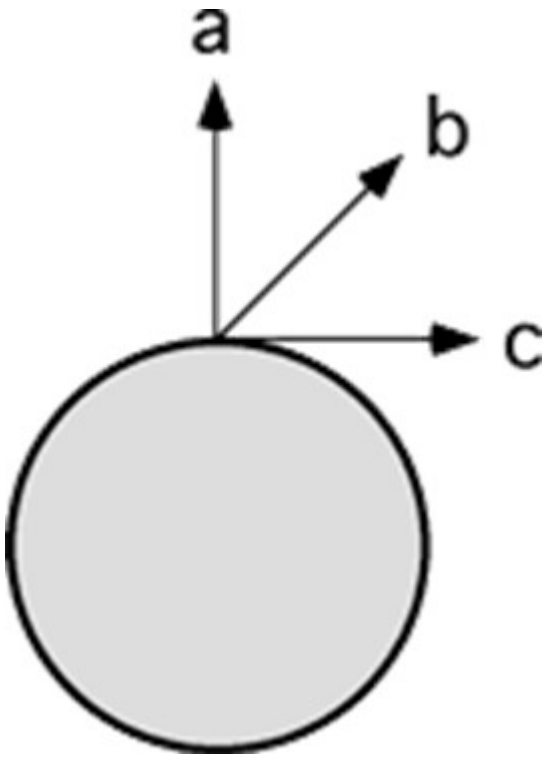
Two masses m_1 and m_2, having $m_1 > m_2$, are launched with the same speed in the direction b. Which mass reaches the greatest height?

1. m_1
2. m_2
3. Both reach the same height.



Assessing to Learn

A mass *m* is launched from the surface of the Earth with speed *v*. The diagram shows three possible launch directions; order them according to the maximum height the projectile will reach, from greatest height to least height.



1. a = b = c
2. a < b = c
3. a < b < c
4. a < c < b
5. a > b = c
6. a > b > c
7. a > c > b
8. b > a > c
9. b > a = c
10. Not enough information.